WORKED SOLUTION

STEP III 2016, Question 9

Mechanics • Simple harmonic motion

STEP III 2016, Question 9

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/16-s3-q9

The key idea

Put the origin at the centre of the triangle and use symmetry. The equilibrium position is the centre, where the three equal tensions cancel. When the particle moves along the line towards P, the springs QX and RX stay mirror images of each other, so their sideways pulls cancel and the motion stays on that line. Finding RX needs the cosine rule. For the period we write down the exact restoring force and then keep only the terms that are linear in the small displacement.

Solution

We model X as a particle on a smooth table and the springs as light springs obeying Hooke's law, so a spring of length r has tension $\frac{\lambda}{l}(r-l)$.

Equilibrium

Let O be the centre of the triangle. Each vertex is at distance

$$c = \frac{2}{3} \times (\text{height}) = \frac{2}{3} \cdot \sqrt{3}\, a = \frac{2a}{\sqrt{3}}$$

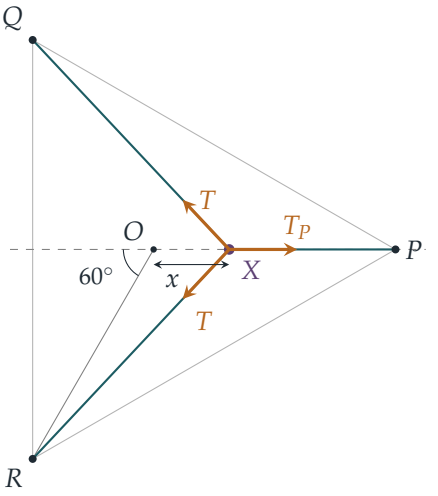
from O, and the lines OP, OQ, OR are at 120° to each other. With X at O, each spring has length c, so all three tensions are equal. Three equal forces at 120° to each other add to zero, so O is an equilibrium position. This is the equilibrium we use. Since $l < \frac{2}{\sqrt{3}}a$, each spring is stretched, with extension

$$\frac{2}{\sqrt{3}}\, a - l,$$

as required.

The tension in RX

Take the x-axis along OP, with P in the positive direction. Let X be at displacement x from O towards P. Then $OX = x$, $OR = c$ and $\angle XOR = 180° - 60° = 120°$.



By the cosine rule in triangle XOR,

$$RX^2 = c^2 + x^2 - 2cx\cos 120° = c^2 + cx + x^2 = \frac{4a^2}{3} + \frac{2ax}{\sqrt{3}} + x^2.$$

So by Hooke's law the tension in RX is

$$T = \frac{\lambda}{l}\left(\sqrt{\frac{4a^2}{3} + \frac{2ax}{\sqrt{3}} + x^2} - l\right),$$

as required. By symmetry in the axis OP, $QX = RX$, so QX has the same tension T.

The equation of motion

Write $r = RX$. The spring PX has length $c - x$, so its tension is $T_P = \frac{\lambda}{l}(c - x - l)$, acting towards P (in the $+x$ direction).

The point R is at x-coordinate $-c\cos 60° = -\frac{c}{2}$, so the displacement from X to R has x-component $-\left(x + \frac{c}{2}\right)$. The force T along XR therefore has x-component $-T\,\dfrac{x + c/2}{r}$. The force from QX has the same x-component, and the two perpendicular components cancel by symmetry. So there is no force across the line OP, and X moves along it.

Ask yourself How do we get a linear equation out of this square root?

First write the exact force, then expand for small x. Resolving along OP,

$$m\ddot{x} = T_P - 2T\,\frac{x + c/2}{r} = \frac{\lambda}{l}\left[(c - x - l) - (2x + c)\left(1 - \frac{l}{r}\right)\right]$$
$$= \frac{\lambda}{l}\left[-3x - l + \frac{l(2x + c)}{r}\right].$$

This equation is exact. Now $r^2 = c^2\left(1 + \frac{x}{c} + \frac{x^2}{c^2}\right)$. Using the binomial expansion and dropping terms in x^2 and higher,

$$\frac{1}{r} \approx \frac{1}{c}\left(1 - \frac{x}{2c}\right), \qquad \frac{2x + c}{r} \approx \left(1 + \frac{2x}{c}\right)\left(1 - \frac{x}{2c}\right) \approx 1 + \frac{3x}{2c}.$$

Substituting,

$$m\ddot{x} \approx \frac{\lambda}{l}\left[-3x - l + l + \frac{3lx}{2c}\right] = -\frac{3\lambda}{l}\left(1 - \frac{l}{2c}\right)x.$$

Since $\frac{l}{2c} = \frac{\sqrt{3}\,l}{4a}$,

$$\ddot{x} \approx -\omega^2 x, \qquad \omega^2 = \frac{3\lambda(4a - \sqrt{3}\,l)}{4mla}.$$

This ω^2 is positive because $\sqrt{3}\,l < 2a < 4a$. So the force is restoring, and for small x the particle performs approximate simple harmonic motion about O with period

$$\frac{2\pi}{\omega} = 2\pi\sqrt{\frac{4mla}{3(4a - \sqrt{3}\,l)\lambda}},$$

as required.

Checks

- At $x = 0$ the exact force is $\frac{\lambda}{l}\left[-l + \frac{lc}{c}\right] = 0$, which agrees with the equilibrium found first.
- Units: λ is a force and l, a are lengths, so $\frac{mla}{a\lambda}$ has units $\mathrm{kg\,m/N} = \mathrm{s}^2$, and the period is a time.
- A second route to ω^2. For a small displacement, a spring contributes stiffness $\frac{\lambda}{l}\cos^2\phi$ from its stretching and $\frac{T_0}{c}\sin^2\phi$ from turning, where ϕ is its angle to OP and $T_0 = \frac{\lambda}{l}(c - l)$ is the equilibrium tension. With $\phi = 0°, 60°, 60°$ the total stiffness is $\frac{3}{2}\cdot\frac{\lambda}{l} + \frac{3}{2}\cdot\frac{T_0}{c} = \frac{3\lambda}{l}\left(1 - \frac{l}{2c}\right)$, as before.
- With $a = \lambda = m = 1$ and $l = 0.8$ the formula gives a period of about 2.838. A computer simulation of the full two-dimensional motion, starting from a small displacement towards P, gives the same value to four significant figures.

Common slips

- Using $\cos 120^\circ = +\frac{1}{2}$, or the wrong angle in the cosine rule. This gives $-\frac{2ax}{\sqrt{3}}$ inside the square root.
- Treating the tension in QX and RX as constant. The change in their *direction* and the change in their *size* both contribute terms linear in x.
- Writing the side of the triangle as a rather than $2a$ when finding the distance from the centre to a vertex.

Beyond STEP

Near any stable equilibrium, the force is approximately linear in the displacement. Here the restoring force is the gradient of the elastic potential energy, and $m\omega^2$ is its second derivative. In two dimensions that second derivative becomes a 2×2 matrix (the Hessian), and its eigenvectors are the *normal modes* of small oscillation. By the threefold symmetry, this system oscillates with the same period in every direction about O, not just along OP. Kleppner and Kolenkow, *An Introduction to Mechanics*, covers small oscillations and coupled systems.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.